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Deformation behaviour of Mg–0.7 wt.% Nd alloy

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Abstract

Deformation behaviour of the Mg–0.7 wt.% Nd alloy was investigated in tensile tests at elevated temperatures. Obtained local maximum in the temperature dependence of the yield stress as well as in the maximum stress indicates a presence of the dynamic strain ageing. The Portevin–Le Chatelier effect was observed at temperatures between 200 and 300 °C. In the microstructure of deformed samples many twins were observed. The twinning mode of deformation decreases with increasing test temperature.
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Keywords: Metals; Casting; Dislocations; Point defects; Strain

1. Introduction

It is well known that rare earths have favourable influence on mechanical properties of magnesium alloys especially at higher temperatures [1,2]. High performance magnesium rare earth alloys offer considerable advantages to the aerospace and racing automotive industry [1]. Key properties of these alloys include high specific strength at room and elevated temperatures, good creep resistance and the excellent castability and damping characteristics. However, despite if use of magnesium rare earth alloys for approaching 50 years, the scientific understanding of the influence of rare earth on mechanical properties is incomplete.

The aim of this work is to investigate the deformation behaviour of the Mg–0.7 wt.% Nd at elevated temperatures and to account for the results obtained.

2. Experimental procedures

Magnesium of commercial purity with 0.7 wt.% Nd was used in this study. Cylindrical specimens with 5 mm of diameter and 25 mm of length for tensile tests were prepared from squeeze cast ingots. Thermal treatment *T*6 (homogenisation at 525 °C for 5 h, then quenching into water 65 °C warm with subsequent precipitation treating for 8 h at 204 °C). Deformation tests were performed in an Instron machine in the temperature interval from the room temperature to 300 °C at an initial strain rate from 5.5×10^{-5} to $1.1 \times 10^{-3}\,s^{-1}$. Microstructure of deformed and undeformed samples was observed by Olympus light microscope.

3. Results and discussion

The Fig. 1 shows a representative microstructure of undeformed samples after *T*6 temperature. Grains in the microstructure are formed by the hexagonal close packed (hcp) solid solution of Nd in Mg with particles inside the grains and decorating the grain boundaries. Solubility of Nd in Mg is small. Park and Wyman [3] placed the maximum at 0.1 at.% at the eutectic temperature. According to [4] the precipitation process occurs in Mg–Nd alloy as follows: solid solution → GP zones → β″ Mg_3Nd cph phase → β′ Mg_3Nd (fcc) → β $Mg_{12}Nd$ phase. The addition of Nd in dilute amounts to Mg results in significant increases on hardening and strength after adequate thermal treatment [5]. This increase in the hardness was reported due to formation of plate-shaped GP zones and precipitates on prismatic planes of the Mg matrix [6]. While structure of the precipitate plates (designated as β″) was commonly reported to be $D0_{19}$ superlattice the habit plane of the precipitate plates has

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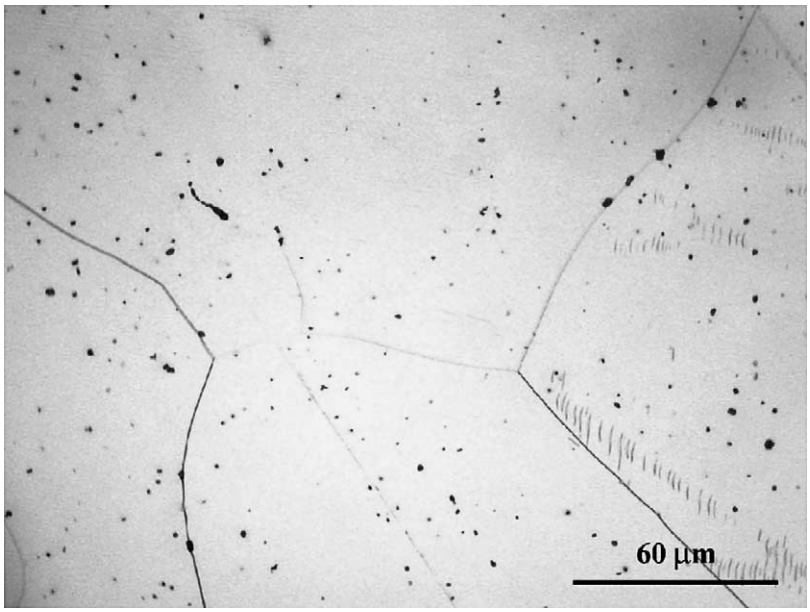


Fig. 1. Microstructure of Mg–Nd alloy after *T*6 temperature.

been reported to be parallel to either $\{1\,1\,\bar{2}\,0\}_\alpha$ or $\{1\,0\,\bar{1}\,0\}_\alpha$ planes of the matrix phase. The GP zones were reported to have a faulted $D0_{19}$ structure and to form on $\{1\,0\,\bar{1}\,0\}_\alpha$ planes of the matrix phase [5,6].

The true stress–true strain curves obtained at various temperatures are introduced in Fig. 2. From Fig. 2 it is obvious that strain hardening was observed in the whole temperature interval studied. Ductility of the alloy is relatively low even at elevated temperatures. Serrated yielding was observed at temperatures 250–300 °C at the strain rate of $\dot{\varepsilon} = 3.3 \times 10^{-4}\,s^{-1}$. Temperature dependences of the yield stress as well as the maximum stress are introduced in Fig. 3. It can be seen that the course of the temperature dependence is non-monotonous. A local maximum in the temperature dependence of both the yield stress and the maximum stress is observed in the temperature interval of 200–250 °C.

Characteristic feature of the microstructure after deformation is a lot of deformation twins (Fig. 4a,b). Twins were observed in whole temperature interval studied, their density decreased with increasing temperature. Twins as the important deformation mode were observed by Máthis et al. in the case of AM60 magnesium alloy [7]. The formation of twins contributes also to deformation. Twinning may reorient the basal planes that become more favourable for slip. Zhang et al. [8] have reported this behaviour. The twinning planes are $\{1\,0\,\bar{1}\,2\}$ and $\{3\,0\,\bar{3}\,4\}$ but at higher temperatures twins were observed in plane $\{1\,0\,\bar{1}\,3\}$, too [9].

Serrated character of stress–strain curves obtained at 200–300 °C can be caused by either twinning or the Portevin–LeChâtelier (PLC) effect. To decide this problem stress–strain curves were performed at 250 °C and various strain rates from 5.5×10^{-5} to $1.1 \times 10^{-3}\,s^{-1}$. The strain rate dependence of the yield stress is introduced in Fig. 5. Negative strain rate sensitivity of this dependence in the strain rate interval from 5.5×10^{-5} to $5.5 \times 10^{-4}\,s^{-1}$ indicates a presence of the unstable

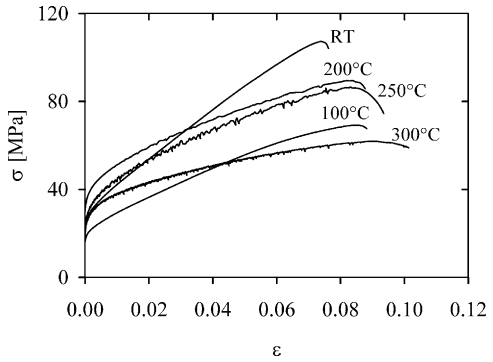


Fig. 2. Stress–strain curves obtained at various temperatures, initial strain rate $3.3 \times 10^{-4}\,s^{-1}$.

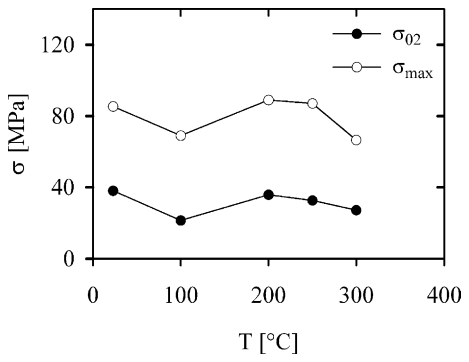


Fig. 3. Temperature dependence of the maximum stress and the yield stress.

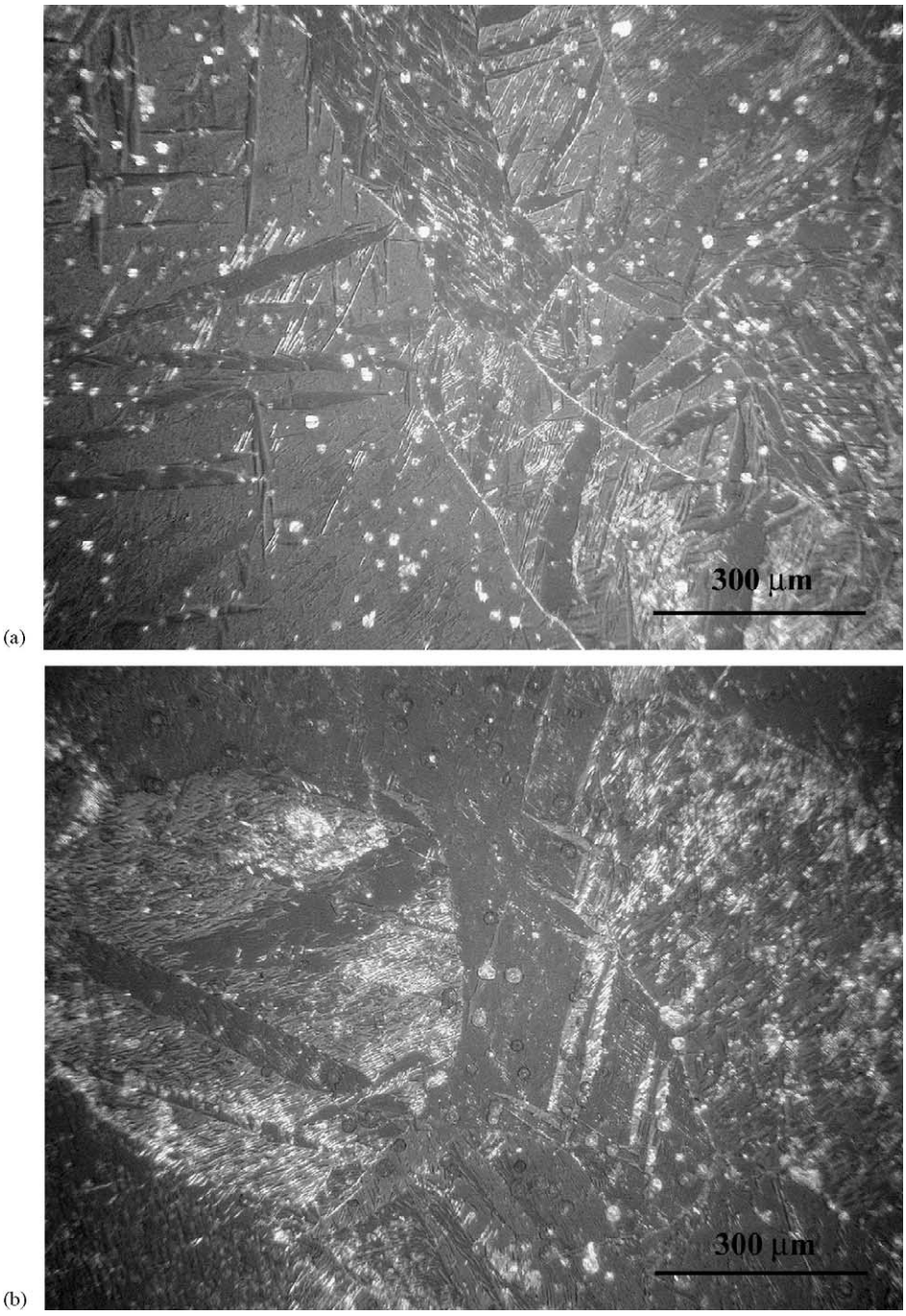


Fig. 4. Twins observed after deformation at (a) 100 °C and (b) 300 °C.

plastic deformation which is manifested by the PLC effect (see Fig. 6). From the microscopic point of view, it is widely accepted that the PLC effect arises from the solute diffusion assisted obstruction of dislocation motion. It appears when the solute mobility is comparable with the dislocation velocity. The solute concentration increases generally around the cores of mobile dislocations exhibiting repeated transitions between the enriched state and the saturated one and vice versa, which corresponds to the slow and fast deformation regime, respectively. The microscopic models of PLC effect were developed in several papers [10–14]. PLC effect requires some critical density of dislocations and critical waiting time for dislocations pinned at obstacles. It is also reason why the PLC effect apears at some critical strain and in the relatively narrow strain rate interval. Movable solute atoms are necessary for the occurrence of the PLC effect. Although the alloy studied obeyed to $T6$ temperature it has very probably sufficient number of solute atoms for the occurence of such mechanism.

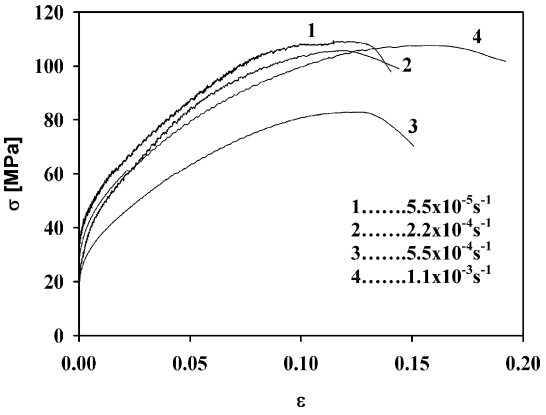


Fig. 5. Stress–strain curves obtained at various strain rates at 250 °C.

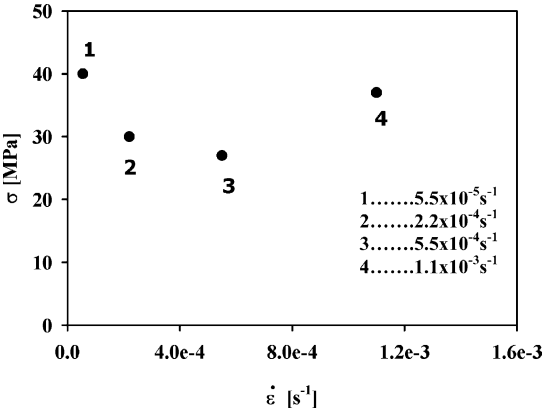


Fig. 6. Strain rate dependence of the yield stress.

4. Conclusions

Deformation behaviour of Mg–0.7Nd alloy was studied at elevated temperatures. Temperature dependence of the yield stress as well as the maximum stress is non-monotonous. Local maximum between 200–250 °C is a consequence of dynamic strain ageing in the alloy. The second demonstration of the strain ageing phenomenon is the presence of the Portevin–LeChâtelier effect which was observed in the temperature interval from 200–300 °C. The negative slope in the strain rate dependence of the yield stress is the evidence of the PLC effect. Free solution atoms in the solid solution are the reason for strain ageing effects. Twinning deformation mode was observed in the whole temperature interval and its influence decreases with increasing test temperature.

Acknowledgements

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